

Magezi — “The Wise One”

Wisdom in your language — A-Level STEM Tutor

Helping rural students master the 2025 Competence-Based Curriculum, one step at a time.

License MIT LLM Groq Llama 70B Frontend Next.js 16 Backend FastAPI KB 4,457 passages

The Problem

Over **70% of Uganda’s A-Level students** attend rural schools where qualified STEM teachers are scarce. The 2025 NCDC Competence-Based Curriculum demands higher-order thinking — but most students have no one to guide them.

Magezi is an AI tutor that speaks the student’s language (Luganda, Swahili, Runyankole, English), follows the official NCDC 2025 syllabus, and teaches step by step.

How It Works

Student asks a question (voice or text, any language)

- Input guardrails (injection + cheating detection)
- Supervisor classifies subject (zero-latency keyword classifier)
- Agentic tutor plans response (complexity, multi-retrieval, tools)
- Curriculum RAG (4,457 passages from 288 official NCDC PDF pages)
- STEM tools inject verified formulas + constants
- Groq Llama 3.3 70B generates pedagogical response (with retry)
- Fallback: structured local response if LLM unavailable
- Markdown rendering (headers, bold, lists, code, tables)
- Faithfulness scoring + citations to exact syllabus pages
- Multi-turn context (5-turn sliding window per session)

Key Features

Feature	Description
Groq LLM	Llama 3.3 70B via Groq — fast inference with auto-retry on 429
4,457 Passages	288 pages of official NCDC 2025 PDFs + structured JSON syllabus
4 Languages	Luganda, Swahili, Runyankole, English — 120-term multilingual bridge
4 Subjects	Physics, Chemistry, Biology, Mathematics (A-Level)
Agentic Tutor	Plans before answering: complexity analysis, multi-retrieval, tool use
STEM Tools	18 physical constants + 47 formulas + safe calculator
Markdown Chat	react-markdown + remark-gfm: headers, bold, lists, code, tables render properly
Grok-Style Actions	Copy, Regenerate, Listen, Thumbs up/down on every response
Voice I/O	Web Speech API input + TTS output + auto-narrate toggle
Multi-Turn Context	5-turn sliding window — follow-ups remember previous discussion
Quiz Mode	“Test me on Newton’s Laws” generates competence-based practice questions
Auth + Credits	Email signup → 50 free credits → BYOK for unlimited
Scroll-to-Bottom	Floating button when scrolled up (Grok-inspired)
Mobile-First	100dvh chat layout, sticky composer, collapsible hero
Offline-Capable	PWA + service worker + offline detection banner
Citations	Every answer cites exact NCDC syllabus page and section
Graceful Fallback	If LLM fails (rate limit), formatted local response with formulas
WCAG 2.2 AA	Skip-link, focus-visible, reduced-motion, contrast 4.7:1

Architecture

```
+-----+
|  Next.js 16 PWA  |
| React 19 + Zustand|
| + TanStack Query |
| react-markdown   |
+-----+-----+
|
+-----+-----+
|  FastAPI Gateway |
| guardrails + auth |
| rate limit + CORS |
+-----+-----+
|
+-----+-----+
|           |           |
+-----+-----+ +-----+-----+ +-----+-----+
| Supervisor | | Tutor Loop | | STEM Tools |
| (classify) | | (plan →   | | 18 constants|
|           | | retrieve → | | 47 formulas |
|           | | synthesise)| | calculator  |
+-----+-----+ +-----+-----+ +-----+-----+
|           |           |
+-----+-----+ +-----+-----+
| Curriculum | | Groq API  |
| RAG (4,457 | | Llama 3.3 |
| passages)  | | 70B + retry|
+-----+-----+ +-----+-----+
```

Knowledge Base

Source	Pages	Passages
Physics.pdf (NCDC 2025)	84	1,321
CHEMISTRY.pdf (NCDC 2025)	70	984
Biology.pdf (NCDC 2025)	44	576
PRINCIPAL MATHS.pdf (NCDC 2025)	90	1,440
Structured JSON syllabus	—	68

Source	Pages	Passages
Total	288	4,457

Quick Start

Prerequisites

- Python 3.11+, Bun (or Node 20+)
- Groq API key (free tier: <https://console.groq.com>)

1. Configure

```
cd Magezi
cp .env.example .env
# Set GROQ_API_KEY in .env
```

2. Ingest knowledge base

```
cd backend
pip install pymupdf
python -m app.indexer
# Extracts 4,389 passages from 4 NCDC PDFs
```

3. Run

```
# Backend (terminal 1)
cd backend
uvicorn app.main:app --port 8802

# Frontend (terminal 2)
cd frontend
bun install && bun run dev --port 3300
```

4. Open

- Frontend: <http://localhost:3300>
- Backend: <http://localhost:8802/docs>

Project Structure

```
Magezi/
├─ README.md / ETHICS.md / DEMO.md / PRESENTATION.md / CLAUDE.md
├─ .env.example
├─ docker-compose.yml
├─ backend/
│   └─ app/
│       ├── main.py          # FastAPI + auth + SSE streaming + quiz
│       └─ service.py        # TutoringService – agentic pipeline +
fallback
│   └─ llm.py                # Groq (Llama 70B) + Claude + retry +
fallback
│   ├── retriever.py         # Hybrid retrieval (Qdrant + keyword)
│   ├── indexer.py           # PDF ingestion (PyMuPDF → chunks)
│   ├── tools.py             # 18 constants + 47 formulas + calculator
│   ├── auth.py              # Email auth + JWT + credits + BYOK
│   ├── guardrails.py        # OWASP LLM01 + cheating detection
│   └─ agents/
│       ├── supervisor.py    # Subject classification
│       └─ tutor_loop.py     # Agentic planning + multi-retrieval
├─ frontend/
│   └─ src/
│       ├── app/page.tsx     # Mobile-first chat + scroll-to-bottom
│       ├── components/
│       │   └─ ChatMessage.tsx # Markdown + copy/retry/listen + expand/
collapse
│       ├── ChatInput.tsx    # Auto-resize textarea + mic + send
│       ├── AuthModal.tsx    # Signup-first + "Continue without account"
│       │   └─ ...
│       ├── store/           # useChatStore + useAuthStore (Zustand)
│       └─ hooks/            # useApi (TanStack) + useSpeech +
useOnlineStatus
│   └─ services/             # voiceService (TTS + AudioRecorder)
├─ knowledge-base/
│   ├── syllabus/           # 4 structured JSON files
│   └─ extracted/           # 4,389 PDF-extracted passages
└─ docs/                   # 4 official NCDC 2025 syllabus PDFs
```

Team

Mpairwe Lauben — Makerere University, Final Year Computer Science

Hackathon Category

Category 3: Economic Empowerment & Education

Built with Groq + Claude | Makerere University Claude Hackathon 2026